

GCSE PAPER 2 PROBLEM SHEET A

HINTS & SOLUTIONS: fifteen problems for Edexcel higher paper two (calculator).

Hints. Give every question your best attempt before looking at the solutions!

1. Directly proportional means $t = ks$ for some constant k . Inversely proportional means $c = k/\sqrt{d}$. (I remember that k is always multiplying: it's $t = k \times s$ and $c = k \times 1/\sqrt{d}$).

3, 12. Use the variable storing feature of your calculator! It's much harder to make a mistake this way.

4. The question is asking for percentage **increase**. How much has each price increased, and what is that increase as a percentage of the original amount?

5. Density = mass / volume. Find the mass of the triangular prism first, then use the fact that this is the same mass as the cuboid and create an equation for l .

6. Remember that the second difference needs to be halved (e.g. if the second difference is 8, then the leading term is $4n^2 + \dots$).

7. Factorise! Difference of two squares?

8. Write the initial number of counters in the bag as some variable. I chose to say there were $10x$ counters in the bag to begin, with $4x$ red and $6x$ blue. Then create and solve a quadratic equation (its not the prettiest, but it is a calculator paper!)

11. Always recall LAV for similar shapes. If x is the length scale factor, then x^2 is the area scale factor, and x^3 is the volume scale factor.

14. When data is grouped, one can use the midpoint of the group as the representative for the entire group. In less cryptic terms: if 7 people are in the 20 – 30 age group, it's reasonable to suggest the sum of all ages is 7×25 .

15. Geometric sequences multiply: if a is the first term, and r is the multiplying factor, then the sequence is $a, ar, ar^2, \dots$ (e.g. 10, 30, 90, 270, $\dots$).

The multiplying factor (common ratio) r is always the same between terms, so you can create an equation to solve by looking at the division of consecutive terms:

$$10, 30, 90, \dots \implies 30/10 = 90/30 \quad (r = 3).$$

Solutions are overleaf.

Solutions. Please use the hints and give each question your best attempt first!

1. As $t = ks$ we have $3.5 = k \times 24.5$ or $k = 3.5/24.5$. When $s = 35$ we have $t = 5$.

(b) As $c = k/\sqrt{d}$ we have $1 = k/\sqrt{9} = k/3$ and thus $k = 3$. When $c = 0.12$ we have $\sqrt{d} = k/0.12 = 3/0.12 = 25$, and if $\sqrt{d} = 25$ then $d = 25^2 = 625$.

2. Let the x axis be temperature and the y axis be sales. Points should be scattered on the graph roughly exhibiting an increase (bottom left to top right). The line of best fit goes through the middle of these points.

Prediction can be done by measuring from the given variable (e.g. 25°C or 100 sales), along to the line of best fit, and then run perpendicular along to the other variable.

3. For $y = x^3 + 4x^2 - 6x - 3$.

x	-3	-2	-1	0	1	2	3
y	24	17	6	-3	-4	9	42

See an online graphing calculator, e.g. <https://www.desmos.com/calculator/u59fb1beae>

4. Chocolate has increased £9 (from 5 to 14) and thus an increase of $9/5 = 1.8 = 180\%$. Sandwiches have increased £4.80 and thus an increase of $4.80/8 = 0.6 = 60\%$.

Since $180 = 3 \times 60$, Bob is correct.

5. Recall $\rho = m/V$, where ρ is density, m is mass, and V is volume.

The volume of the triangular prism is $15 \times 12 = 180\text{cm}^3$. Thus $m = 0.8 \times 180 = 144\text{g}$.

This is the same mass as the cuboid. It has volume $4 \times 5 \times \ell = 20\ell\text{cm}^3$, but also $V = m/\rho = 144/0.72 = 200\text{cm}^3$. So $20\ell = 200$ and thus $\ell = 10$.

6. The first difference is 7, 11, 15, 19 and the second difference 4, 4, 4, 4. So the leading order term is $2n^2$ (don't forget to half it!)

Comparing $2n^2$ to the original sequence yields

Sequence :	4, 11, 22, 37, 56
$2n^2$:	2, 8, 18, 32, 50
Diff :	2, 3, 4, 5, 6.

The difference is another sequence with n th term $n + 1$. Therefore the full expression for the sequence is $2n^2 + n + 1$ (and one should check candidate values of n with your calculator).

(b) The 100th term is $2 \times (100^2) + 100 + 1 = 20101$.

7. Factorise the expression as

$$\frac{2x^2 - 5x - 3}{x^2 - 9} \div \frac{4x^2 - 1}{2x^2 + 5x - 3} = \frac{(2x+1)(x-3)}{(x+3)(x-3)} \div \frac{(2x+1)(2x-1)}{(2x-1)(x+3)},$$

which after cancelling like terms becomes

$$\frac{2x+1}{x+3} \div \frac{2x+1}{x+3} = \frac{2x+1}{x+3} \times \frac{x+3}{2x+1} = 1.$$

8. It is very helpful to define the number of counters in the bag. I chose to say there are $10x$ counters in the bag, of which $4x$ are red and therefore $6x$ are blue (since the probability of a red counter is 0.4).

On the first draw taking a red counter has probability 0.4. On the second draw the probability becomes $(4x-1)/(10x-1)$ because there is one less red counter (and therefore one less counter overall). Then

$$0.4 \times \frac{4x-1}{10x-1} = \frac{13}{85} \implies 34(4x-1) = 13(10x-1)$$

which can be solved with $85 \times 0.4(4x-1) = 13(10x-1)$, thus $136x - 34 = 130x - 13$ or $6x = 21$. There were originally $6x$ blue counters, i.e. there are 21 blue counters originally in the bag (of the 35 total, of which 14 were red).

9. Compound interest builds each year. A 3.95% rate is multiplication by 1.0395, so

$$4500 \times 1.0395 \times 1.0395 \times 1.0395 = 4500 \times 1.0395^3 = 5054.59,$$

so Toby has earned interest of £554.59.

10. For an upper bound on the number of video files we assume the memory stick is at its largest capacity and the video files are as small as they could be. Thus assume the capacity is 64.5gb and each file is 4.15gb, meaning one can store $64.5/4.15 = 15.54 = 15$ full video files.

11. The area scale factor from A to B is $96/54 = 16/9$. Using the LAV: x - x^2 - x^3 rule, the length scale factor is $4/3$ and thus the volume scale factor is $64/27$. So the volume of shape B is $324 \times 64/27 = 768\text{cm}^3$.

12. For $y = 2x^2 - 4x - 1$.

x	-1	0	1	2	3	4
y	5	-1	-3	-1	5	15

(b) See <https://www.desmos.com/calculator/u59fb1beae> . (c) Answers within a reasonable tolerance of -0.58 and 2.58 would be accepted.

13. Let $x = 0.2\dot{4}\ddot{5} = 0.24545\dots$. Then $10x = 2.4545\dots$ and $1000x = 245.4545\dots$. So $990x = 243$ and $x = 243/990 = 27/110$.

14. First find estimates for the total speed in each group by taking the midpoint, then add these up and divide by 60. So

$$\frac{(35 \times 12) + (45 \times 22) + (55 \times 18) + (70 \times 8)}{60} = 49.3\text{mph.}$$

15. The common ratio between terms is equal, so equate

$$\frac{18}{x-14} = \frac{2x+14}{18},$$

which becomes

$$324 = (2x+14)(x-14) \implies 2x^2 - 14x - 520 = 0.$$

This simplifies to $x^2 - 7x - 260 = 0$ which can be solved with the quadratic formula (or through your calculator) to find $x = 20$ or $x = -13$. The corresponding common ratios are $r = 3$ or $r = -2/3$.